LOAN APPROVAL PREDICTION

Applied Data Analytics & Machine Learning Project

Author	Abdulkadir Abdullahi Mohamed
Institution	Dhambaal Institute — Batch BDA01
Dataset	Public Loan Prediction dataset (614 rows; 13 columns before cleaning)
Models	Logistic Regression · Decision Tree · K-Nearest Neighbours
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Project purpose
Explore whether historical home-loan approval outcomes can be predicted from applicant information, compare three classification models, and document the statistical findings and limits of the analysis.

Portfolio project • Educational analysis • Not for automated lending decisions

1. Business Problem & Research Question

Manual review of home-loan applications can be time-consuming and may lead to inconsistent decisions. This project investigates whether patterns in historical application records can help predict the recorded approval outcome.

Research question: Can applicant information such as income, credit history, loan amount and personal characteristics predict whether an application was historically approved?

Problem type: Binary classification. The target variable is `loan_status`, encoded as 1 = approved and 0 = rejected.

2. Dataset & Data Quality

The notebook uses a public educational Loan Prediction dataset with 614 applications and 13 original columns. The ID field is removed before modeling, leaving 12 columns before feature engineering. The target distribution is 422 approved and 192 rejected applications (about 68.7% approved).

Dataset check	Observed result
Rows and original columns	614 rows × 13 columns
Target distribution	422 approved; 192 rejected
Missing values	149 missing cells across 7 columns
Rows with at least one missing value	134 rows (about 22%)
Most-missing feature	credit_history: 50 missing values
Duplicate rows	0 found after dropping the ID field

The notebook cites public Kaggle mirrors of the Loan Prediction dataset. For reproducibility, select and document one exact CSV version, check its usage terms, and rerun the notebook using that file. The CSV was not included in the files supplied for this revision.

3. Statistical Analysis & EDA

The exploratory analysis uses frequency counts, cross-tabulations, approval proportions, distributions, and a correlation heatmap. These methods describe relationships in the sample; they do not establish causation.

Key descriptive findings

• Credit history has the strongest observed association with the recorded outcome: 79.6% of applications with credit_history = 1 were approved, compared with 7.9% where credit_history = 0. Missing values are excluded from this descriptive crosstab.
• Semiurban applications had a 76.8% approval rate, compared with 61.5% for rural applications and 65.8% for urban applications.
• The income distributions for approved and rejected applications overlap substantially.
• The numeric correlation between credit_history and loan_status is about 0.56; applicant income and loan amount are correlated at about 0.57.

4. Data Cleaning & Preprocessing

Structural cleaning removes the ID field, standardizes dependents values by converting “3+” to 3, and maps the target labels to numeric values. Missing values are not filled across the full dataset before splitting.

Leakage-safe process

The data is split into a training set of 491 applications and a held-out test set of 123 applications using stratification. Median imputation is applied to loan amount and loan term; most-frequent imputation is applied to dependents and credit history; categorical fields are imputed and one-hot encoded. Income variables are passed through, and total_income is engineered as applicant income plus co-applicant income.

Preprocessing is implemented in scikit-learn pipelines and fitted on training data. The KNN cross-validation process refits preprocessing within each training fold, avoiding the leakage that would occur if test-fold information influenced imputation, encoding, or scaling.

Methodology clarification

The submitted notebook uses Logistic Regression with max_iter=1000 and default regularization settings. The Decision Tree uses max_depth=4, specified in advance for interpretability; that depth is not cross-validated. KNN evaluates odd k values from 1 to 29 using 5-fold stratified cross-validation on the training set, selecting k by mean cross-validation accuracy. The report intentionally describes these choices exactly as implemented.

5. Models & Evaluation

Three classification models are trained on the same training/test split: Logistic Regression, Decision Tree and K-Nearest Neighbours. The evaluation includes accuracy, precision, recall, F1-score and confusion matrices. The baseline predicts “approved” for every test application.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	86.18%	84.00%	98.82%	90.81%
Decision Tree	84.55%	83.00%	97.65%	89.73%
KNN (k = 21)	81.30%	79.25%	98.82%	87.96%
Baseline: approve all	69.11%	69.11%	100.00%	81.73%

Precision, recall and F1-score in this table use the notebook's positive class: approved (1). They should not be read as measures of how well the model detects rejected applications.

6. Interpreting the Errors

The Logistic Regression confusion matrix shows 22 rejected applications correctly identified as rejected, 16 rejected applications incorrectly predicted as approved, 1 approved application incorrectly predicted as rejected, and 84 approved applications correctly identified as approved. Thus, rejected-class recall is 22/38 = 57.9%, despite the high recall for approved applications.

For the Decision Tree, the reported test-set classification report shows 21 of 38 rejected applications identified and 83 of 85 approved applications identified. For KNN, 16 of 38 rejected applications were identified and 84 of 85 approved applications were identified.

7. Key Findings

• Logistic Regression achieved the highest recorded accuracy and approved-class F1-score among the three models on this test split.
• The Decision Tree is close in overall accuracy, and its feature-importance output assigns about 0.7448 importance to credit_history.
• KNN selected k = 21 using 5-fold cross-validation accuracy on the training set and had the lowest held-out accuracy of the three models.
• All three models exceeded the 69.11% accuracy baseline, but the baseline illustrates why accuracy alone is insufficient when the target is imbalanced.
• Credit history appears highly influential in this dataset, but missing values in this feature make imputation consequential.

8. Limitations & Responsible Use

This project predicts past approval decisions, not whether a borrower will repay a loan. Historical labels can encode prior institutional bias. The dataset is small, contains missing values, and includes gender and marital status as predictors. Credit history is both highly predictive and partly missing. The model also misses a meaningful share of historically rejected applications.

The project should therefore be presented as an educational classification and statistical analysis exercise. It is not suitable for real lending decisions or automated approval/rejection. Any practical application would require exact data provenance, robust external validation, class-specific error analysis, fairness assessment, governance, and meaningful human oversight.

9. Conclusion

The recorded results suggest that historical loan-approval labels can be predicted better than the approve-all baseline on this particular split. Logistic Regression is the strongest model by the reported overall accuracy and approved-class F1-score, but its rejected-class recall is only 57.9%. The main value of this project is demonstrating a structured analytical workflow: inspect and clean data, explore patterns, prevent preprocessing leakage, compare models, interpret errors, and communicate limitations honestly.

10. Reproducibility Checklist

1. Download the exact permitted dataset version and place it beside the notebook as dataset.csv.
2. Install packages listed in requirements.txt (scikit-learn 1.2 or newer is needed for sparse_output=False).
3. Run every notebook cell from top to bottom in a clean kernel.
4. Confirm that the row counts, EDA results, cross-validation-selected k and test metrics match the report.
5. If any result differs, update both the notebook and this report together before publication.

Dataset source cited in the submitted materials: Kaggle Loan Prediction dataset. Verify the exact CSV version and applicable terms before redistribution. This report was revised to align its stated methodology and metrics with the submitted notebook outputs.